Unit- III

Time and Work: Calculate Time to Complete Work by 2 or More People, Equations Based Time and Work Problems, Efficiency Based Time and Work Problems, Calculate Time When Efficiency is Given in Percentage, Calculate Time When Workers Leave in Between, Share of Salary Based on Work.

Time and Work

IMPORTANT FACTS AND FORMULAE

1. If A can do a piece of work in n days, then A's 1 day's work $= \frac{1}{n}$.

2. If A's 1 day's work $= \frac{1}{n}$, then A can finish the work in n days.

3. If A is thrice as good a workman as B, then :

   Ratio of work done by A and B = 3 : 1.

   Ratio of times taken by A and B to finish a work = 1 : 3.

Ex. 1. Worker A takes 8 hours to do a job. Worker B takes 10 hours to do the same job. How long should it take both A and B, working together but independently, to do the same job ? (IGNOU, 2003)

Solution- A's 1 hour's work = 1/8

B's 1 hour's work = 1/10

(A+B)'s 1 hour's work= (1/8 + 1/10) = 9/40

Both A and B will finish the work in 40/9

= 4(4/9) days

Ex. 2. A and B together can complete a piece of work in 4 days. If A alone can complete the same work in 12 days, in how many days can B alone complete that work?

(Bank P.O. 2003)

Solution- (A+B)'s 1 day's work=1/4

A's 1 day's work= 1/12

B's 1 day's work= (¼ -1/12)= 1/6

B can complete the work in 6 days.

Q. A do the work in 10 days, B do the work in 15 days then together they required how many days?

Solution- Method1- A=10, B=15

A=1/10, B=1/15 => A+B= 1/10 + 1/15 =1/6 =6

one day work

Method2- A=10

B=15

30
LCM
(Total
Work)

3

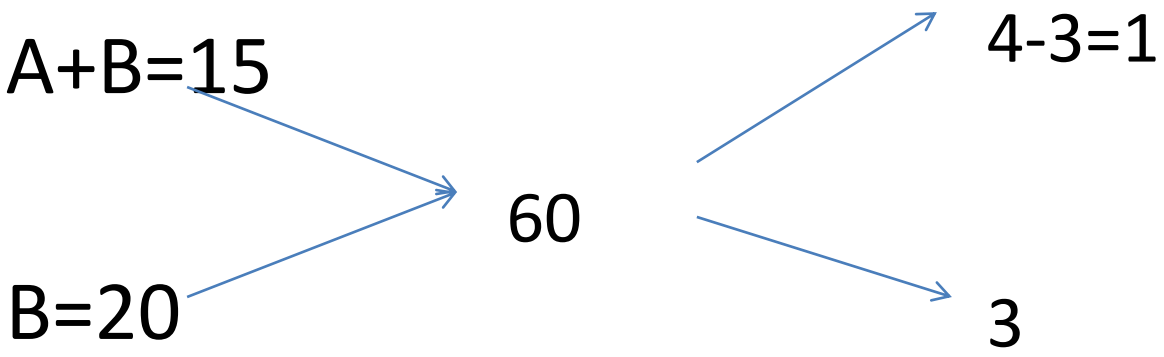
2

5

30/5= 6 days

Q. A and B together do the work in 15 days, B alone can do the work in 20 days, then A alone can do the work in how many days?

Solution-



60/1= 60 days

Ex. 3. *A can do a piece of work in 7 days of 9 hours each and B can do it in 6 days of 7 hours each. How long will they take to do it, working together* $8\frac{2}{5}$ *hours a day?*

Solution- A can complete the work in 7*9= 63h

B can complete the work in 6*7= 42h

A's 1 hour work=1/63, B's 1 hour work=1/42

(A+B)'s 1 hour work= 1/63 + 1/42 = 5/126

Both finish the work in 126/5 hours

Number of days of 8(2/5) hrs each=
(126/5)*(5/42)= 3 days

4. A, B and C can complete a piece of work in 24, 6 and 12 days respectively. Working together, they will complete the same work in : (C.B.I. 2003)

(a) $\frac{1}{24}$ day (b) $\frac{7}{24}$ day (c) $3\frac{3}{7}$ days (d) 4 days

Solution- 1/24 + 1/6 +1/12 = 7/24

completion= 24/7 = 3 (3/7)

(Hotel Management

6. A man can do a piece of work in 5 days, but with the help of his son, he can do it in 3 days. In what time can the son do it alone ? (S.S.C. 2004)

(a) $6\frac{1}{2}$ days (b) 7 days (c) $7\frac{1}{2}$ days (d) 8 days

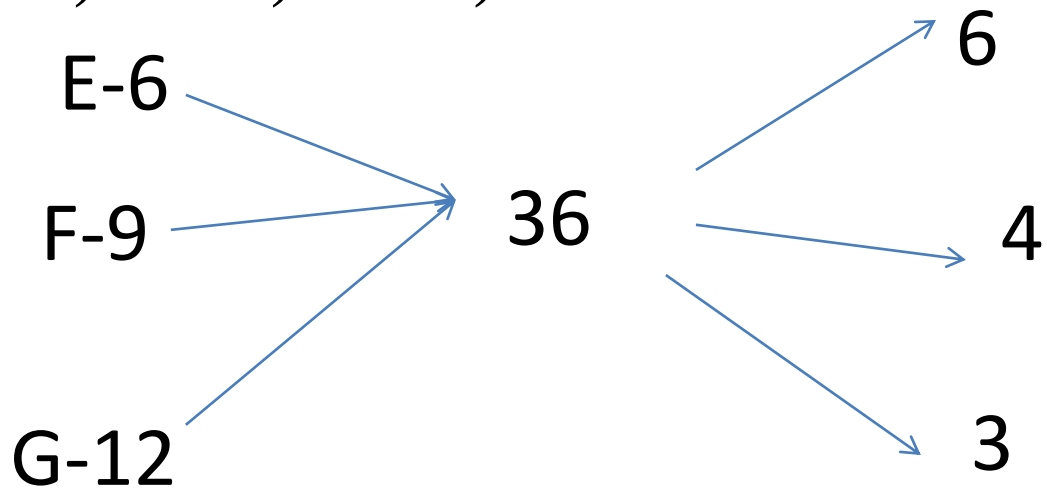
Solution- 1/3 – 1/5 = 2/15

15/2= 7 (1/2)

- E can complete a job in 6 days. F can complete it in 9 days. G can complete it in 12 days. E and F started the job. After 3 days, G took over and complete it. Find the time for which G worked(in days). (IBPS RRB PO 2017)

a) 2 b) 3 c) 4 d) 1 f)6

Solution-

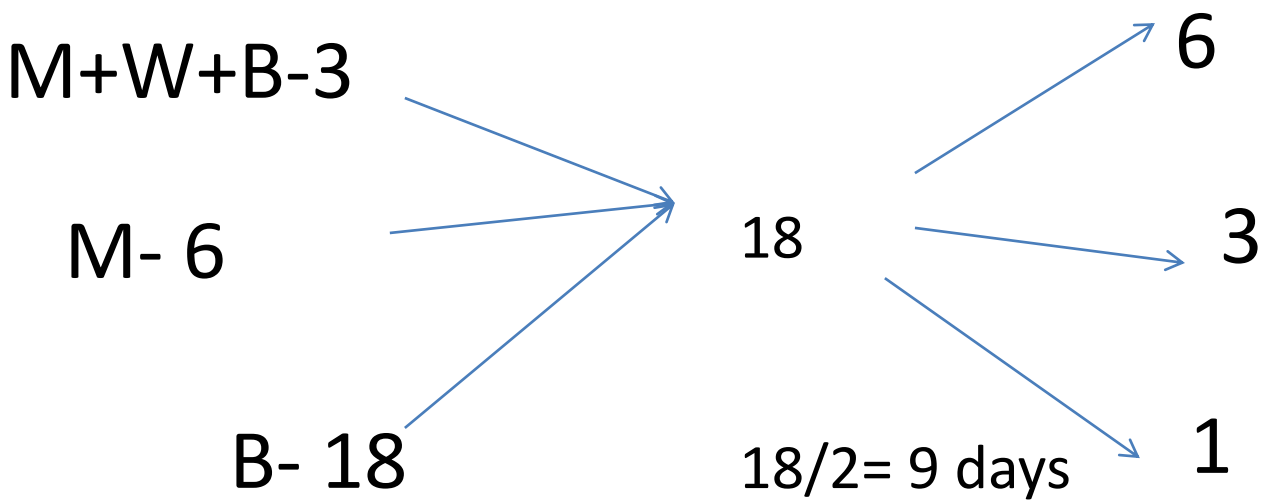


10x3= 30 36-30= 6/3= 2 days

* A man, a woman and a boy can together complete a piece of work in 3 days. If a man can do it in 6 days and a boy alone in 18 days, how long will a woman alone take to complete the work?

a) 9 b)21 c) 24) 27

Solution-



- A is thrice as efficient as B. A started working and after 4 days he was replaced by B. B then worked for 15 days and left. If A and B together finished 75% of the total work, in how many days B alone can finish the whole work? **(SBI PO 2016)**

Solution-

A : B

3 : 1

4x3+ 1x15=27

75%= 27

100%= (27/75) *100= 36 days

Ex. 7. A can do a piece of work in 80 days. He works at it for 10 days and then B alone finishes the remaining work in 42 days. In how much time will A and B, working together, finish the work ?

Solution- A's work done in 10 days =(1/80)x10= 1/8

Remaining work = (1- (1/8))= 7/8

Now 7/8 work is done by B in 42 days.

Whole work will be done by B in 42x(8/7)= 48 days

A's day's work = 1/80 , B's 1 day's work = 1/48

(A+B)'s 1 day's work= (1/80 + 1/48)=1/30

Both will finish the work in 30 days.

- George is 2/3rd as efficient as smith and smith is 3/4th as efficient as John. George working alone is what fraction of all of them working together. (**TCS**)

G : S

G:S:J

2:3:4

= 2/(2+3+4)

= 2/9

2/3 : 1

2 : 3

S : J

¾ : 1

3 : 4

- A and B completed a work together in 5 days. Had A worked at twice the speed and B at half the speed, it would have taken them four days to complete the job. How much time would it take for A alone to do the work? (**TCS**)

Ans- (A+B)'s 1 day's work = 1/5---------(1)

2A+B/2=1/4----------(2)

2x(2)- (1)

4A+B-A-B= ½-1/5

3A= 3/10

A= 1/10

A alone would take 10 days

- George can do some work in 8 hours, Paul can do the same work in 10 hours while Hillari can do the same work in 12 hours. All the three of them start working at 9 a.m. while George stops works at 11 a.m. and remaining two complete the work. Approximately at what time will the work be finished (**TCS**)

a) 12 pm b) 1 pm c) 1:30 pm d) 2 pm

Solution- G- 8

P- 10

H- 12

120
-74

46/22=
2

15

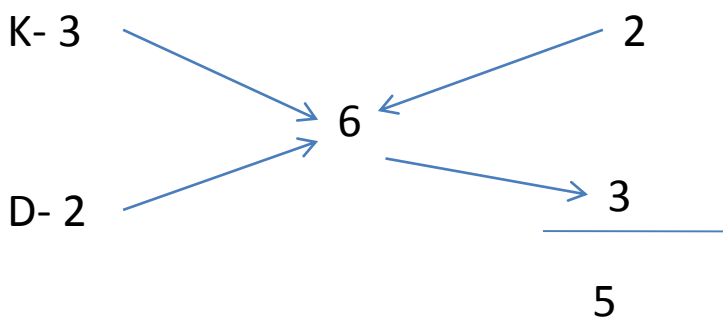
12

10

37

2x37= 74= (G+P+H)

* Kim can do a work in 3 days while David can do it in 2 days. Both of them finish the work together and get Rs. 150/- what is the share of Kim?



150/ 5 = 30

K – 2x30 = Rs. 60
D- 3x30 = Rs. 90

Unit-III

Pipes and Cisterns: Important Facts and Formulae, Calculate Time Taken to Fill a Tank By 2 or More Pipes, Calculate Time Taken to Fill a Tank With Leakage, Equations Based Pipes and Cistern Problems, Calculate Time Taken When Pipes Are Opened For Different Periods, Calculate Number of Pipes.

Pipes and Cisterns

IMPORTANT FACTS AND FORMULAE

1. **Inlet** : A pipe connected with a tank or a cistern or a reservoir, that fills it, is known as an inlet.

   Outlet : A pipe connected with a tank or a cistern or a reservoir, emptying it, is known as an outlet.

2. (*i*) If a pipe can fill a tank in x hours, then :

   part filled in 1 hour $= \frac{1}{x}$.

   (*ii*) If a pipe can empty a full tank in y hours, then :

   part emptied in 1 hour $= \frac{1}{y}$.

   (*iii*) If a pipe can fill a tank in x hours and another pipe can empty the full tank in y hours (where $y > x$), then on opening both the pipes, the net part filled in 1 hour $= \left(\frac{1}{x} - \frac{1}{y}\right)$.

   (*iv*) If a pipe can fill a tank in x hours and another pipe can empty the full tank in y hours (where $x > y$), then on opening both the pipes, the net part emptied in 1 hour $= \left(\frac{1}{y} - \frac{1}{x}\right)$.

Ex. 1. *Two pipes A and B can fill a tank in 36 hours and 45 hours respectively. If both the pipes are opened simultaneously, how much time will be taken to fill the tank?*

Sol. Part filled by A in 1 hour $= \frac{1}{36}$; Part filled by B in 1 hour $= \frac{1}{45}$.

Part filled by (A + B) in 1 hour $= \left(\frac{1}{36} + \frac{1}{45}\right) = \frac{9}{180} = \frac{1}{20}$.

Hence, both the pipes together will fill the tank in 20 hours.

Ex. 2. *Two pipes can fill a tank in 10 hours and 12 hours respectively while a third pipe empties the full tank in 20 hours. If all the three pipes operate simultaneously, in how much time will the tank be filled ?*

Sol. Net part filled in 1 hour $= \left(\frac{1}{10} + \frac{1}{12} - \frac{1}{20}\right) = \frac{8}{60} = \frac{2}{15}$.

$\therefore$ The tank will be full in $\frac{15}{2}$ hrs = 7 hrs 30 min.

Ex. 3. *If two pipes function simultaneously, the reservoir will be filled in 12 hours. One pipe fills the reservoir 10 hours faster than the other. How many hours does it take the second pipe to fill the reservoir ?*

Solution- Let the reservoir be filled by first pipe in x hrs

Then, Second pipe will fill it in (x+10) hours

$1/x + 1/(x+10) = 1/12$

$x+10+x/x(x+10)= 1/12$

$X^2 – 14x -120=0$

$(x-20) (x+6) = 0$

$x=20$ [neglecting the –ve value of x]

So the second pipe will take (20+10) hrs i.e. 30 hrs to fill reservoir

4. A cistern has two taps which fill it in 12 minutes and 15 minutes respectively. There is also a waste pipe in the cistern. When all the three are opened, the empty Cistern is full in 20 minutes. How long will the waste pipe take to empty the full cistern?

Sol. Work done by the waste pipe in 1 minute

$$= \frac{1}{20} - \left(\frac{1}{12} + \frac{1}{15}\right) = -\frac{1}{10}$$ [– ve sign means emptying]

$\therefore$ Waste pipe will empty the full cistern in 10 minutes.

Ex. 5. *An electric pump can fill a tank in 3 hours. Because of a leak in the tank, it took $3\frac{1}{2}$ hours to fill the tank. If the tank is full, how much time will the leak take to empty it ?*

Sol. Work done by the leak in 1 hour $= \left[\frac{1}{3} - \frac{1}{\left(\frac{7}{2}\right)}\right] = \left(\frac{1}{3} - \frac{2}{7}\right) = \frac{1}{21}$.

$\therefore$ The leak will empty the tank in 21 hours.

Ex. 6. ***Two pipes can fill a cistern in 14 hours and 16 hours respectively. The pipes are opened simultaneously and it is found that due to leakage in the bottom it took 32 minutes more to fill the cistern. When the cistern is full, in what time will the leak empty it ?***

Sol. Work done by the two pipes in 1 hour $= \left(\frac{1}{14} + \frac{1}{16}\right) = \frac{15}{112}$.

$\therefore$ Time taken by these pipes to fill the tank $= \frac{112}{15}$ hrs $= 7$ hrs 28 min.

Due to leakage, time taken = 7 hrs 28 min + 32 min = 8 hrs

$\therefore$ Work done by (two pipes + leak) in 1 hour $= \frac{1}{8}$.

Work done by the leak in 1 hour $= \left(\frac{15}{112} - \frac{1}{8}\right) = \frac{1}{112}$.

$\therefore$ Leak will empty the full cistern in 112 hours.

Ex. 7. ***Two pipes***

1. Two pipes A and B can fill a tank in 20 and 30 minutes respectively. If both the pipes are used together, then how long will it take to fill the tank ? (M.A.T. 2003)

(a) 12 min (b) 15 min (c) 25 min (d) 50 min

Solution- Part filled by A in 1 minutes=1/20

Part filled by B in 1 minutes=1/30

Part filled by (A+B) in 1 minutes=1/20 +1/30=1/12

3. A tap can fill a tank in 6 hours. After half the tank is filled, three more similar taps are opened. What is the total time taken to fill the tank completely ?

(*a*) 3 hrs 15 min (*b*) 3 hrs 45 min (*c*) 4 hrs (*d*) 4 hrs 15 min

(S.S.C. 2003)

3. Time taken by one tap to fill half the tank = 3 hrs.

Part filled by the four taps in 1 hour $= \left(4 \times \frac{1}{6}\right) = \frac{2}{3}$.

Remaining part $= \left(1 - \frac{1}{2}\right) = \frac{1}{2}$.

$\therefore \quad \frac{2}{3} : \frac{1}{2} :: 1 : x \quad$ or $\quad x = \left(\frac{1}{2} \times 1 \times \frac{3}{2}\right) = \frac{3}{4}$ hrs *i.e.*, 45 mins.

So, total time taken = 3 hrs 45 min.

4. A water tank is two-fifth full. Pipe A can fill a tank in 10 minutes and pipe B can empty it in 6 minutes. If both the pipes are open, how long will it take to empty or fill the tank completely ? **(Bank P.O. 1999)**

(*a*) 6 min. to empty (*b*) 6 min. to fill (*c*) 9 min. to empty

(*d*) 9 min. to fill (*e*) None of these

4. Clearly, pipe B is faster than pipe A and so, the tank will be emptied.

Part to be emptied $= \frac{2}{5}$.

Part emptied by (A + B) in 1 minute $= \left(\frac{1}{6} - \frac{1}{10}\right) = \frac{1}{15}$.

$\therefore \quad \frac{1}{15} : \frac{2}{5} :: 1 : x \quad$ or $\quad x = \left(\frac{2}{5} \times 1 \times 15\right) = 6$ min.

So, the tank will be emptied in 6 min.

5. Pipe A can fill a tank in 5 hours, pipe B in 10 hours and pipe C in 30 hours. If all the pipes are open, in how many hours will the tank be filled ? (C.B.I. 1997)

(a) 2 (b) 2.5 (c) 3 (d) 3.5

Solution-1/5+1/10+1/30 = 1/3

6. Pipes A and B can fill a tank in 5 and 6 hours respectively. Pipe C can empty it in 12 hours. If all the three pipes are opened together, then the tank will be filled in :

(a) $1\frac{13}{17}$ hours (b) $2\frac{8}{11}$ hours (c) $3\frac{9}{17}$ hours (d) $4\frac{1}{2}$ hours

(Bank P.O. 2002)

Solution- 1/5 + 1/6 – 1/12 = 17/60

Tank will be full in 60/17 hours= 3(9/17)

13. A tank is filled in 5 hours by three pipes A, B and C. The pipe C is twice as fast as B and B is twice as fast as A. How much time will pipe A alone take to fill the tank?

(*a*) 20 hrs (*b*) 25 hrs (*c*) 35 hrs

(*d*) Cannot be determined (*e*) None of these **(Bank P.O. 2003)**

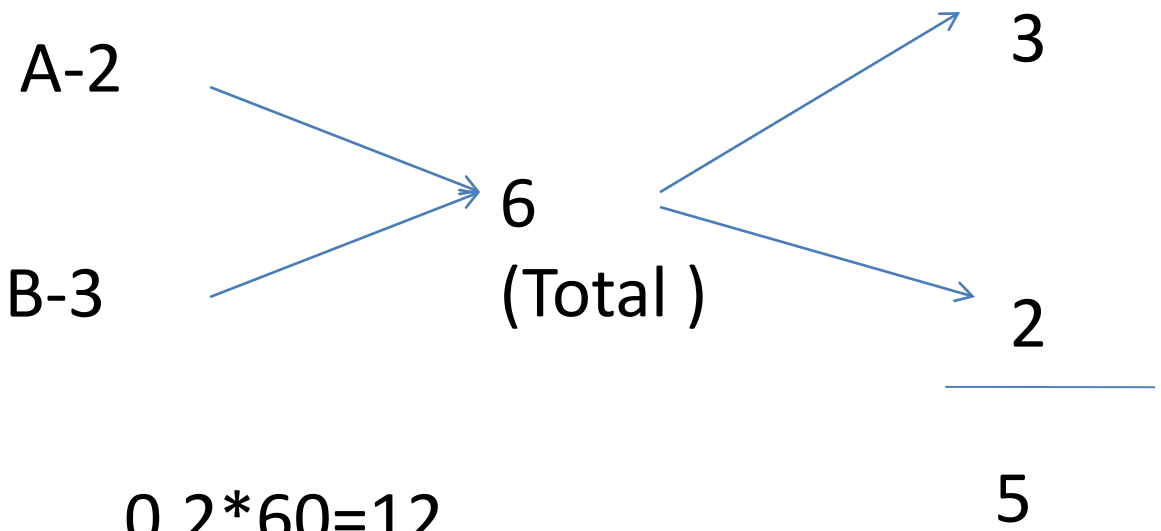
Solution- suppose pipe A alone can take x hours to fill the tank

Then Pipe B and C will take x/2 and x/4 hours to fill the tank = 1/x + 2/x + 4/x = 1/5

7/x= 1/5

x= 35 hrs.

- Two pipes can fill a tank in 2 hours and 3 hours respectively. If they will open together in an empty tank, then how much time they take to fill the tank?



6/5= 1.2 0.2*60=12

1 hour 12 minute

- A tap fills a tank in 8 hours but due to hole in a tank, it takes 2 hours more. How much time the hole will take to empty the tank?

a) 15hrs b)20hrs c) 24hrs d) 40hrs

A- 8

B-10

40
(Total)

5

–

4

1

Solve = 40/1= 40 hours

- A pipe can fill a water tank 3 times faster than in comparison of other pipe. If they both filled the tank in 36 minutes, then how much time will it take to fill the tank by pipe with slower speed?

  a) 1hr 21 min b) 1hr 48 min
  c) 2 hr 21 min d) 2hr 24 min

Solution- A : B

1 : 3 = 4x36 =144/60= 2.4

= .4x 60= 24min

- Two taps P and Q can fill a water tank in 12 and 15 minutes respectively. They were opened together but after 3 minutes tap P was closed. How much more time tap Q take to fill the tank?

Solution-

P-12

Q-15

60

27

33/4=

8(1/4)

5

+

4

9